MINCHAN KIM

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EDUCATION

Qingdao Daewon School, Qingdao, China, *Class of 2027* Aug 2024 to June 2027 (expected)

- Unweighted GPA 3.984 / 4.0. Intended study: artificial intelligence and related fields.
- AP exams planned for May 2027: Calculus BC, Physics C (Mechanics; Electricity and Magnetism), Physics 1, Computer Science A.

HONORS AND AWARDS

All awards below were issued by Qingdao Daewon School and confirmed in the school's record of awards.

- Research and science: 1st Place, Excellent Research Paper, Science and Technology (Dec 2025); 1st Place, Science Fair, High School Division (Apr 2026).
- Subject Excellence Award: Precalculus (Dec 2025, Jul 2026); Creative Writing (Dec 2025).
- Academic Improvement Award: Economics (Dec 2024), English Academic Reading (Dec 2024, Jul 2026), Economics 10 and English Language and Composition (Jul 2025), Physics (Dec 2025), Chinese 11 (Jul 2026).
- Contests: 1st Place, English Speaking Contest (Nov 2024); 2nd Place, school-wide Korean Literature Writing Contest, prose (May 2025); 1st Place, Social Studies Project Creativity (Jun 2025); Popularity Award, Project Creativity, Chinese (Oct 2025).
- Chinese Month: 2nd Place, Poster Making (Sep 2024); 3rd Place, Paper-Weaving and Chinese Calligraphy Contest (Sep 2025); 3rd Place, Poster Making (Mar 2026).
- Performance Excellence Award: Point of View club (Jul 2025); COSMOS (Jul 2026). Certificate of Participation, DAEMUN II (Sep 2025).
- Sports (1st Place): men's basketball and futsal (May 2025); kickball (Jun 2026).

RESEARCH AND PROJECTS

F-22-Inspired RC Aircraft: Design and Flight Stability Sep 2025 to Jan 2026

- Designed the airframe in FreeCAD, built and flight-tested a foam-board aircraft, and wrote a 22-page research paper relating design variables to flight stability. Won 1st Place in the school research paper competition.

Mechanism of Autopilot (school physics paper)

- Analyzed a control-systems paper and an Airbus patent on autopilot disengagement alerts; drew the autopilot-to-manual transition logic and compared illustrative P, PI and PID error curves. Literature-based, written with teacher guidance; no flight or hardware testing.

Environmental ESG Initiatives at Embry-Riddle Aeronautical University (school environmental science paper)

- Wrote a short case study from the university's public materials on flight training, sustainable aviation fuel, campus facilities and research funding. No original data.

Tern, flight-search web app (flytern.site) Mar 2026 to present

- Built with Next.js, the Duffel API and React Three Fiber in English, Korean, Japanese and Chinese; compares fares, duration, layovers and airport weather, with a rule-based schedule-risk score. Checkout runs through an external partner; no revenue.

Automated Drone-Tracking Turret (school science fair)

- Built a camera-tracking rig with Python, OpenCV and Arduino-driven servos; tested YOLOv5 detection and a color-tracking dual-turret demonstration.

EXPERIENCE

JSL Logistics, *Intern* Aug 2026 to present

- Develop staff tools and the public website, including a document-based support chatbot that retrieves source documents, photo-assisted damage review that staff verify, and a 3D cargo-loading planner that checks dimensions, rotation, stacking support and payload limits.

Mintech, Asan, South Korea, *Intern, Manufacturing Technology* Jul 20 to Aug 19, 2026

- Assisted with manufacturing OLED deposition equipment; learned the structure and process flow of sputtering thin-film deposition equipment; assisted with dimensional checks using vernier calipers and micrometers, equipment checks and workplace organization; completed 4M, 3-jeong 5S and workplace safety onboarding.

Korea Aerospace University Aerospace and Space Camp, Goyang, South Korea Aug 3 to 7, 2026

- Trained on flight training devices (FTD), air traffic control (ATC), CubeSat design and autonomous drones; recognized for strong performance in the FTD and ATC simulations. Visited airline headquarters and completed evacuation slide and safety training.

Shandong University Chinese Language and Culture Workshop, Weihai, China Jul to Aug 2025

- Practiced Mandarin daily and joined cultural activities with an international group.

LEADERSHIP AND ACTIVITIES

COSMOS Aero-Engineering Club, *Co-founder; President since Mar 2026* Sep 2025 to present

- Co-led a rocket thrust-stand project through two static-fire tests; after the first test failed to save data, corrected

the recording sequence and captured a continuous thrust record reaching about 190 N and a total impulse of 270.6 N·s.
- Direct Project AEROVIEW, a suction wind tunnel designed in FreeCAD and OpenSCAD to compare NACA 0012, 2412 and 4412 airfoils; found and corrected a design-to-fabrication dimensional error by direct measurement.
- Coordinate TVC-Hopper, which combines reinforcement-learning simulation, avionics and CAD for thrust-vector control (calibration and physical validation in progress).

DAEMUN III (Daewon Model UN), *Deputy Head of Technology* 2026

- Co-developed the conference platform with Junwon Kim: public site, AI assistant, organizer dashboard, and delegate document submissions with version history, team notifications and committee-level publishing. Built the assistant's first backend, admin tools and chat widget, and expanded its FAQ to 103 entries.
- After the assistant stopped during a school-wide demonstration, called a team meeting; the team moved to a gateway-based design with a fallback provider and full-site context, and the next 51 logged queries recorded no upstream errors.

Student Council, Qingdao Daewon School, *Planning Department Vice President* 2025 to Mar 2026

- Helped plan a joint sports day for the domestic and international divisions; ran the live scoreboard at the orientation event and helped with equipment setup, rehearsals and cleanup.

Orchestrators (formerly Mechanics Club), *Vice President* Mar 2026 to present

- Student AI development team of six. Contributed to Agent Device, a palm-sized companion device for AI agents: assigned PCB design verification for the v0.3 keyring board (layout script-generated before review); wrote its order review, checked the module footprint and pin assignments against the datasheet, flagged a 1.6 mm board versus 0.8 mm case thickness mismatch, and wrote a pad-and-pin audit tool with regression tests. The order decision is pending; the board is not yet assembled and firmware has not started.
- Submitted a researcher and creator workflow for the team's Slack agent (pull request under review).
- As Mechanics Club, designed and tested a 4-bit ripple-carry adder from logic-gate ICs and LED indicators; first sketched the flight-booking concept that became Tern.

DAEMUN II (Daewon Model UN), *Delegate for Slovakia, UNCSTD* 2025

- Spoke in debate on drone weaponization and drafted clauses proposing an international drone-forensics hub.

Support Group, *Group Leader (Grade 11)* 2024 to 2026

- Joined four semesters of the school's peer-mentoring program; led the group in Grade 11.

PSYCHOFISH peer-support club; Point of View club (PR team) 2024 to 2025

- Helped run an anonymous peer-support channel and designed recruitment posters; wrote rules and posters for a school-wide Manito event.

SERVICE

- School service, Nov 2024 to Jun 2026: classroom equipment management, school cleaning, orientation assistant, student council event support, and graduation ceremony support including production of the ceremony video.
- Volunteer at Qingdao Tonghang House, a welfare facility for people with intellectual disabilities, Oct to Nov 2025.

OTHER

- Airbus Aerospace Virtual Work Experience, Springpod online programme (Mar 2026); Claude Development and AI Fluency, Anthropic online course (Jul 2026).
- Participant, Goyang Mayor's Cup three-cushion billiards tournament (Jul 2026).

SKILLS

Python, TypeScript, Next.js, React, PostgreSQL, MongoDB, APIs, OpenCV, Arduino, FreeCAD, OpenSCAD. Languages: Korean, English, Mandarin.